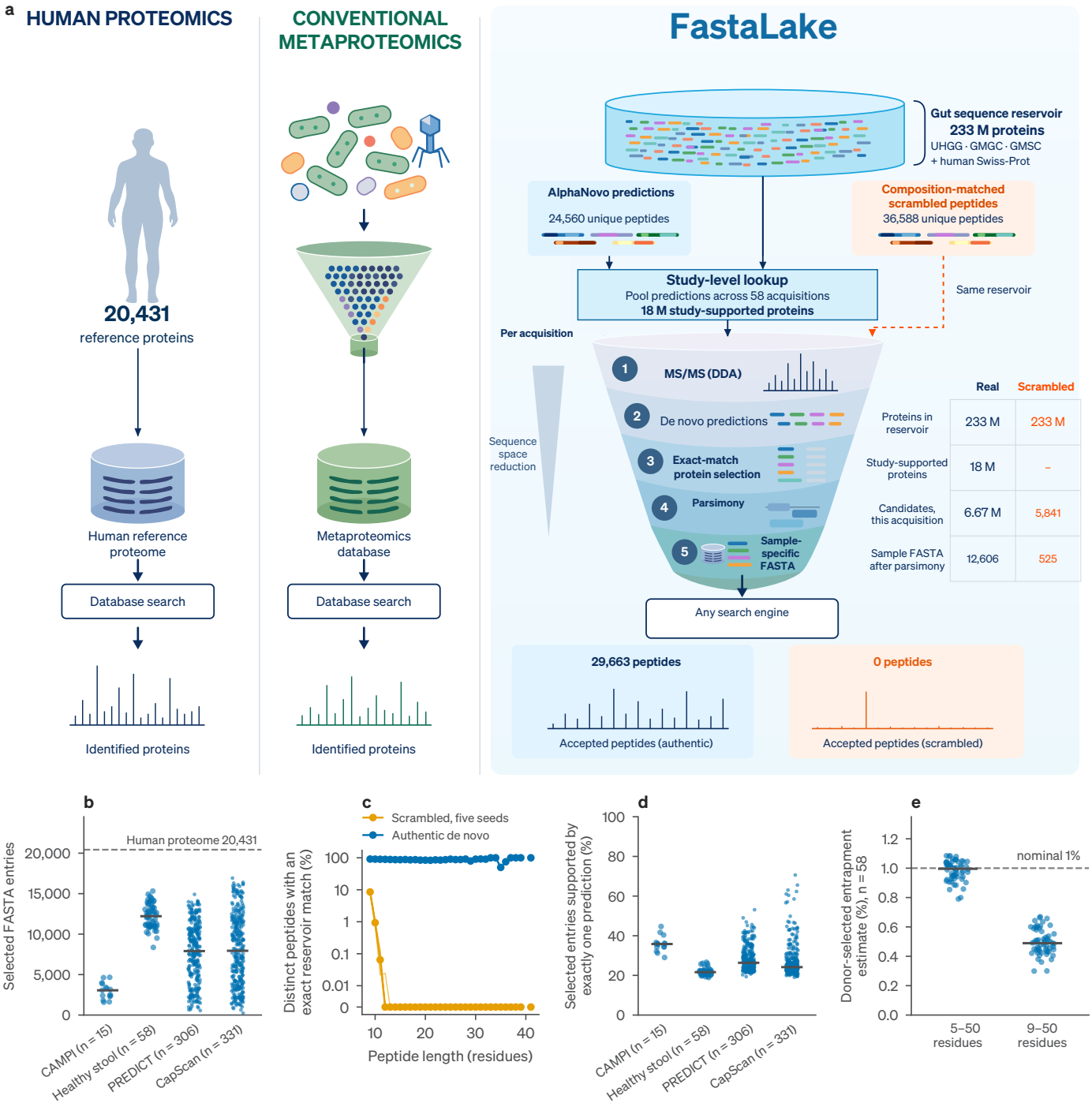

a
HUMAN PROTEOMICS
20,431
reference proteins
Human reference proteome
Database search
Identified proteins
CONVENTIONAL METAPROTEOMICS
Metaproteomics database
Database search
Identified proteins
FastaLake
Gut sequence reservoir
233 M proteins
UHGG · GMGC · GMSC
+ human Swiss-Prot
AlphaNovo predictions
24,560 unique peptides
Composition-matched scrambled peptides
36,588 unique peptides
Same reservoir
Study-level lookup
Pool predictions across 58 acquisitions
18 M study-supported proteins
Per acquisition
1 MS/MS (DDA)
2 De novo predictions
3 Exact-match protein selection
4 Parsimony
5 Sample-specific FASTA
Sequence space reduction
Any search engine
Real
Scrambled
Proteins in reservoir
233 M
233 M
Study-supported proteins
18 M
–
Candidates, this acquisition
6.67 M
5,841
Sample FASTA after parsimony
12,606
525
29,663 peptides
Accepted peptides (authentic)
0 peptides
Accepted peptides (scrambled)
b
Selected FASTA entries
Human proteome 20,431
20,000
15,000
10,000
5,000
0
CAMPI (n = 15)
Healthy stool (n = 58)
PREDICT (n = 306)
CapScan (n = 331)
c
Scrambled, five seeds
Authentic de novo
Distinct peptides with an exact reservoir match (%)
100
10
1
0.1
0.01
0
10
20
30
40
Peptide length (residues)
d
Selected entries supported by exactly one prediction (%)
100
80
60
40
20
0
CAMPI (n = 15)
Healthy stool (n = 58)
PREDICT (n = 306)
CapScan (n = 331)
e
Donor-selected entrapment estimate (%), n = 58
1.2
1.0
0.8
0.6
0.4
0.2
0.0
nominal 1%
5–50 residues
9–50 residues